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Q1) a) Define Data Mining

Data mining is the process of extracting useful patterns, knowledge, and information from large amounts of data

Any Two Data Mining Tasks

- * Classification - classifies data into predefined classes
- * Clustering - Groups similar data without predefined labels

Steps in KDD Process

- * Data cleaning
- * Data integration
- * Data selection
- * Data transformation
- * Data mining

Role of Data processing

- * Removes errors and missing values
- * Improves data quality
- * Reduces processing time

b) Data cleaning

Data cleaning is the process of removing incorrect, incomplete or inconsistent data.

Two data quality issues

- * Missing values
- * Noisy data

Handling Missing values

* Replace with mean/median

* fill manually

* ignore records

Ex Age = ?, replace with average age

salary = ₹ 10,00,000 among ₹ 20,000

②

a) data integration

Data integration combines data from multiple sources into one dataset

challenges

1. data redundancy

2. data inconsistency

Handling inconsistency

* standardize formats

* correct conflicting values

b) covariance Analysis

covariance measures how two variables change together

Two covariance forms

1. Positive covariance

2. Negative covariance

③

a) given

A = (2, 3, 5, 4, 6)

B = (5, 8, 10, 11, 14)

As stock A increases, stock B also increases.

a) Equal - Frequency Binning

Data .

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

Total = 25 values

Bin 1 : 13, 15, 16, 16, 19

Bin 2 : 20, 20, 21, 22, 22

Bin 3 : 25, 25, 25, 30, 33

Bin 4 : 33, 35, 35, 35, 36

Bin 5 : 40, 45, 46, 52, 70

b) data discretization

data discretization convert continuous values into intervals

Equal-width binning

* Same Interval width

* Example:

0 - 20

21 - 40

41 - 60

Equal width

same interval size

simple

May have uneven data

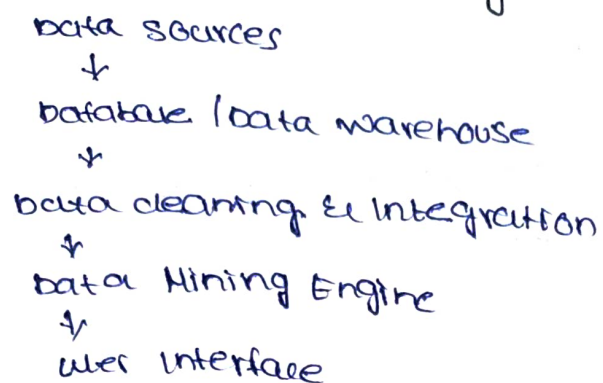
Equal frequency

same no of values

Handles skewed data better.

Balanced bins

b) Architecture of Data Mining System



Functions

data sources → store raw data

database → organizes data

preprocessing → cleans data

a) given

$$x = \{4, 5, 6, 9\} \quad y = \{8, 3, 7, 5, 6\}$$

$$\text{mean of } x = 6.2 \quad \text{mean of } y = 5.8$$

covariance

$$= -0.85$$

b) major steps in preprocessing

* data cleaning

* data integration

* data transformation

* data reduction

Pipeline

Missing values → Remove duplicates → handle outliers → standardize formats → Transform data → final dataset.